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JAVA PROJECT REPORT ON

INVENTORY PURCHASE ORDER MANAGEMENT SYSTEM

Submitted by

VISHNU PRIYAN G

25BCS061

Under the Guidance of

Dr. BOOPALAN SUNDARAMOORTHY

Technical Head

Technical Department, KG College of Arts and Science

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DECLARATION

I hereby declare that the project entitled "**INVENTORY PURCHASE ORDER MANAGEMENT SYSTEM"** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of **Dr. BOOPALAN SUNDARAMOORTHY**,Technical Head,Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

VISHNU PRIYAN G

CERTIFICATE

This is to certify that the project entitled **"INVENTORY PURCHASE ORDER MANAGEMENT SYSTEM"** is a **bonafide work** carried out by **VISHNU PRIYAN G, I - BSC CS 'A' [2025 - 2028]**, **Department Of Computer Science** , **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Head,KGCAS)

_____

Head of the Department
(Technical Head,KGCAS)

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ABSTRACT

The Inventory Purchase Order Management System is a software application developed to manage inventory and purchase orders efficiently. In many organizations, inventory tracking and purchase order handling are done manually, which can lead to errors, delays, and mismanagement of stock.

This system automates the process of managing inventory, generating purchase orders, and tracking stock levels. It allows users to add products, monitor available stock, and create purchase orders when stock reaches a minimum level.

The application ensures accuracy, reduces manual effort, and improves efficiency in inventory control. It is developed using core programming concepts and can be implemented as a console-based or GUI-based application.

The main objective of the system is to provide a simple, reliable, and user-friendly solution for managing inventory and purchase operations.

1. INTRODUCTION

Inventory management plays a vital role in any business organization. It involves tracking products, maintaining stock levels, and ensuring that goods are available when needed. In traditional systems, inventory management is often done manually using registers or spreadsheets, which can lead to errors, data loss, and inefficiency.

Managing purchase orders manually is also a challenging task. Businesses need to keep track of suppliers, product availability, and order status. Without an automated system, this process becomes time-consuming and difficult to manage.

With the advancement of technology, automated inventory systems have become essential for improving efficiency and accuracy. A computerized Inventory Purchase Order Management System helps businesses manage their stock effectively, track inventory in real-time, and generate purchase orders automatically when required.

This system simplifies the process of inventory management by allowing users to enter product details, update stock, and monitor inventory levels. It reduces manual effort and ensures accurate record maintenance. The system also demonstrates the practical application of programming concepts in solving real-world problems.

1.1 Objective

The main objective of the Inventory Purchase Order Management System is to develop a simple and efficient application that automates inventory tracking and purchase order management using programming concepts.

The specific objectives of this project are:

- To design and develop a system for managing product inventory
- To automate stock tracking and updates
- To generate purchase orders when stock levels are low
- To reduce manual errors in inventory management
- To provide a user-friendly interface for easy operation
- To maintain accurate records of products and stock
- To demonstrate the use of programming concepts such as classes, loops, and conditional statements
- To improve efficiency and reduce time consumption in managing inventory

1.2 Overview

The Inventory Purchase Order Management System is a simple application designed to manage inventory and purchase operations efficiently. It provides a digital platform where users can store product details, monitor stock levels, and generate purchase orders when required.

The system allows users to input product information such as product ID, product name, quantity, and supplier details. It continuously tracks stock levels and alerts the user when the quantity falls below a predefined limit. Based on this, the system generates purchase orders automatically or manually.

The application processes all data internally and provides results in a clear and structured format. It eliminates the need for manual calculations and record maintenance. The system can be implemented as a console-based application or extended into a graphical user interface (GUI) for better usability.

1.3 Features

The Inventory Purchase Order Management System includes the following features:

- **Product Management** – Allows users to add, update, and delete product details
- **Stock Tracking** – Monitors inventory levels in real-time
- **Purchase Order Generation** – Creates purchase orders when stock is low
- **Low Stock Alert** – Notifies users when inventory reaches minimum level
- **Supplier Management** – Stores supplier information
- **Report Generation** – Displays inventory and purchase details
- **Error Reduction** – Minimizes manual errors in stock handling
- **User-Friendly Interface** – Easy to understand and operate
- **Fast Processing** – Quickly updates and displays results

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

In many businesses, inventory and purchase orders are managed manually. This approach has several disadvantages:

- Difficulty in maintaining accurate stock records
- Errors in purchase order calculations
- Time-consuming process
- Risk of stock shortages or overstocking
- Lack of proper tracking system
- Data loss due to manual handling

Manual systems increase workload and reduce efficiency. Therefore, an automated system is required to manage inventory and purchase orders effectively.

The Inventory Purchase Order Management System provides a computerized solution to overcome these problems.

2.2 Project Overview

The Inventory Purchase Order Management System is designed to manage inventory and automate purchase operations. It allows users to maintain product details, track stock levels, and generate purchase orders.

The system includes the following functionalities:

- Adding and managing product details
- Updating stock levels
- Generating purchase orders
- Monitoring inventory status
- Displaying reports

This project is implemented using core programming concepts and can be extended into a real-time application.

2.3 Scopes and Limitations

Scope

The scope of the Inventory Purchase Order Management System includes the following:

- Used in small shops and businesses
- Helps in managing inventory efficiently
- Automates purchase order generation
- Provides accurate stock information
- Demonstrates programming concept

The system can be used in:

- Retail stores
- Warehouses
- Manufacturing industries
- E-commerce businesses
- Hospital and pharmacies

Limitations

- Works only with user input (no automation with real-time data)
- Limited to basic inventory operations
- No database integration
- No advanced reporting features
- No online/cloud support

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before implementation. In this phase, the system is divided into modules and the workflow is defined to ensure efficient development.

The Inventory Purchase Order Management System is designed as a simple application where users can manage product details, track inventory levels, and generate purchase orders automatically. The system is divided into modules to perform specific tasks efficiently. Each module is responsible for handling a particular function such as product management, stock tracking, purchase order processing, and report generation.

3.1 Modules

The main modules of the Inventory Purchase Order Management System are explained below:

1. Product Management Module

This module handles product-related operations. It allows users to add, update, and delete product details.

Functions of this module include:

- Accept product ID and name
- Store product information
- Update product details
- Delete products

2. Inventory Module

This module manages stock levels of products.

Functions of this module include:

- Track available stock
- Update quantity after purchase or usage
- Monitor low stock levels

3. Purchase Order Module

This module handles purchase order creation.

Functions of this module include:

- Generate purchase orders
- Store supplier details
- Manage order status

4. Result Display Module

This module displays system outputs.

The result includes:

- Show product details
- Display stock status
- Generate reports

It ensures that the output is easy to understand and visually clear for users.

5. Exit Module

This module allows safe termination of the program.

Functions include:

- Providing exit option to the user
- Terminating the program safely
- Ensuring no data loss during exit

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps from understanding the problem to designing, implementing, testing, and documenting the system.

For the Inventory Purchase Order Management System, a structured approach is followed to ensure accurate stock levels of products.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is analyzed to understand the requirements of the system based on stock level needs.

The main problems identified in manual Inventory Purchase Order Management System are:

- Manual tracking errors
- Time consumption
- Poor record management

Based on these issues, the system requirements are defined to develop an automated application that simplifies purchase order.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.The system is divided into modules such as:

- Product management
- Order handling
- Inventory tracking

The program flow and structure are designed so that users can easily interact with the system and obtain accurate GPA and CGPA results efficiently.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language. The following steps were followed during development:

- Creation of product class
- Implementation of menu-driven interface
- Development of the addproduct()
- Use of FileWriter to store product data
- Handling user inputs and displaying the output

Core Java concepts such as classes, objects, loops, conditional statements, and methods were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results. Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Add Product	Product stored in ArrayList and file
View Products	Displays all added products
Empty List	Shows "No products!"
File Writing	Data saved in products.txt
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system successfully manages inventory and purchase orders with accurate results.

The results are demonstrated through program execution and screenshots showing the input and output of the system.

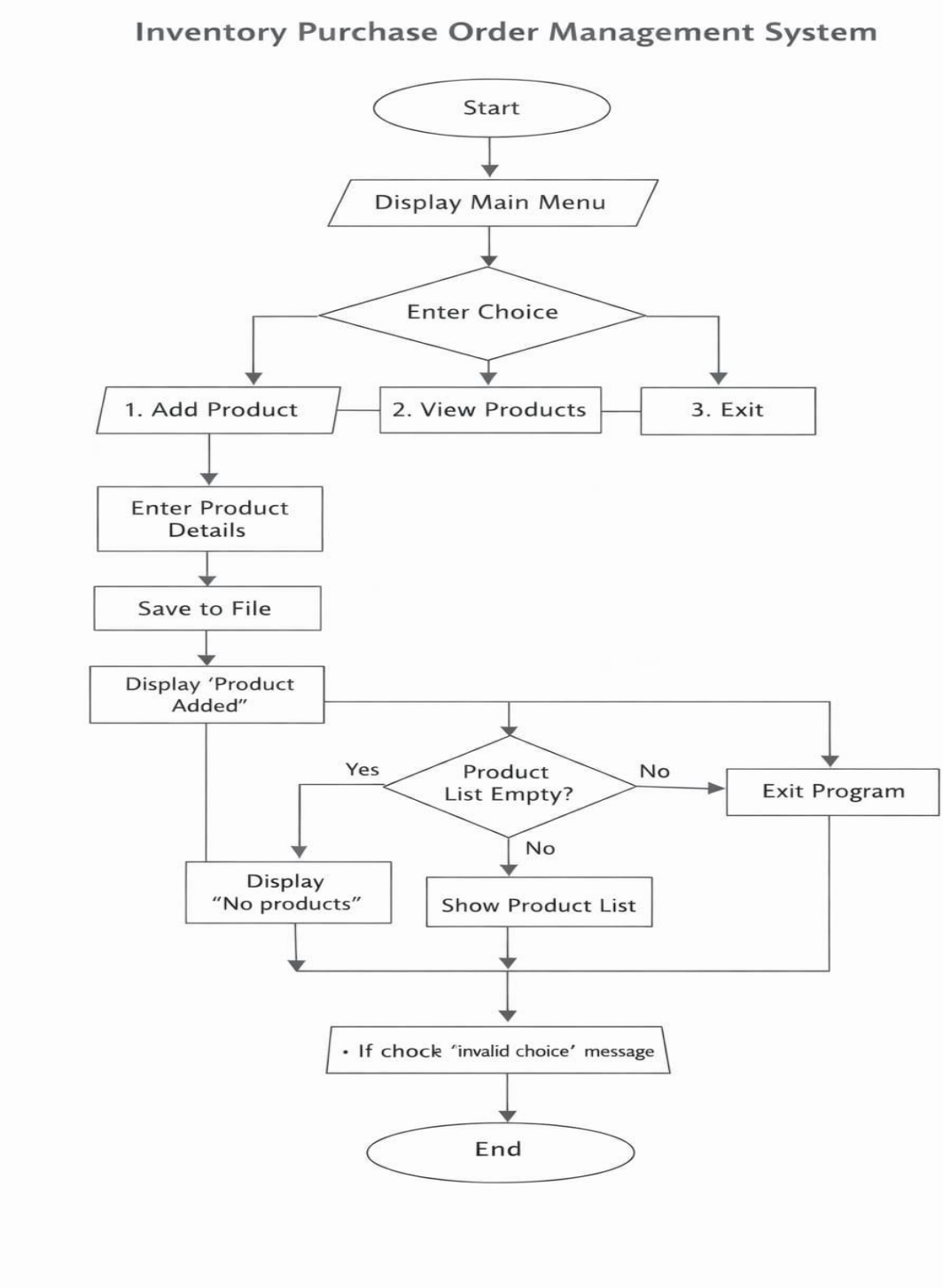
Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the Inventory Purchase Order Management System:

- Start the program.
- Initialize an empty ArrayList to store product details
- Display the main menu with options:
    - Add Product
    - View Products
    - Exit
- Enter user choice, if the choice is Add Product :
    - Enter product ID.
    - Enter product name.
    - Enter product quantity.
    - Create a new product object using entered details.
    - Add the product object to the ArrayList.
    - Display "Product Added" message
- If the choice is View Products:
    - Display product ID, name, and quantity.
- If the choice is Exit:
    - Display exit message.
    - Terminate the program.
- If the choice is invalid, display "Invalid choice" message.
- Repeat the process until the user chooses to exit.
- End the program.

4.7. Flowchart



5. SOURCE CODE

```
import java.util.*;
import java.io.*;

class Product {
   int id;
   String name;
   int quantity;

   Product(int id, String name, int quantity) {
      this.id = id;
      this.name = name;
      this.quantity = quantity;
   }

   void display() {
      System.out.println(id + " | " + name + " | Qty: " + quantity);
   }
}

public class InventorySystem {

   static ArrayList<Product> products = new ArrayList<>();
   static Scanner sc = new Scanner(System.in);

   public static void main(String[] args) {

      while (true) {
         System.out.println("\n===== INVENTORY SYSTEM =====");
         System.out.println("1. Add Product");
         System.out.println("2. View Products");
         System.out.println("3. Exit");

         int choice = sc.nextInt();

         try {
            switch (choice) {
               case 1:
                  addProduct();  // now handled safely
                  break;
               case 2:
                  viewProducts();
                  break;
```

```
                case 3:
                    System.exit(0);
                default:
                    System.out.println("Invalid choice");
            }
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}

static void addProduct() throws IOException {

    System.out.print("Enter ID: ");
    int id = sc.nextInt();
    sc.nextLine(); // fix input bug

    System.out.print("Enter Name: ");
    String name = sc.nextLine();

    System.out.print("Enter Quantity: ");
    int qty = sc.nextInt();

    products.add(new Product(id, name, qty));

    FileWriter fw = new FileWriter("products.txt", true);
    fw.write(id + " " + name + " " + qty + "\n");
    fw.close();

    System.out.println("Product Added!");
}

static void viewProducts() {
    System.out.println("\n--- Product List ---");

    if (products.isEmpty()) {
        System.out.println("No products!");
        return;
    }

    for (Product p : products) {
        p.display();
    }
}
}
```

6. INPUT AND OUTPUT SCREENS

```
===== INVENTORY SYSTEM ====
1. Add Product
2. View Products
3. Exit
Enter choice: 1
Enter ID: 101
Enter Name: Rice Bag
Enter Quantity: 50
Product Added!
Enter choice: 1_
```

```
===== INVENTORY SYSTEM ====
1. Add Product
2. View Products
3. Exit
Enter choice: 2
--- Product List ---
ID      | Name       Qty
-------------------------
101     | Rice Bag | Qty: 50
102     | Sugar Pack | Qty: 100
Enter choice: 1_

1. Add Product
2. View Products
3. Exit
Enter choice: _
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The Inventory Purchase Order Management System was developed to simplify and automate the process of managing product inventory and purchase operations. The system allows users to enter product details such as product ID, name, and quantity, and efficiently stores and displays the inventory information.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, file handling, and basic input/output operations were used. The system ensures accurate storage of product details and enables easy retrieval of inventory data, providing reliable results for users.

The system was tested with different inputs to verify its correctness and efficiency. The results show that the system reduces manual errors in inventory management and saves time, making it easier to maintain and track product details effectively.

7.2 Conclusion

The Inventory Purchase Order Management System was developed to simplify and automate the process of managing product inventory and purchase operations. The system allows users to enter product details such as product ID, name, and quantity, and efficiently stores and displays the inventory information.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, file handling, and basic input/output operations were used. The system ensures accurate storage of product details and enables easy retrieval of inventory data, providing reliable results for users.

The system was tested with different inputs to verify its correctness and efficiency. The results show that the system reduces manual errors in inventory management and saves time, making it easier to maintain and track product details effectively.

8. REFERENCES

The following resources were used during the development of this project:

Books

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